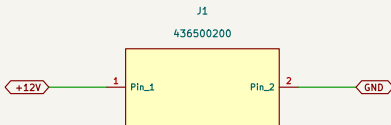
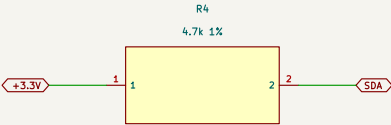
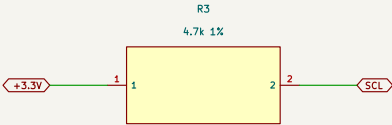
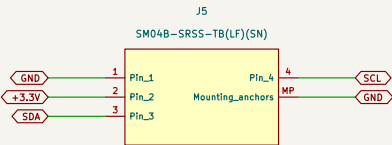
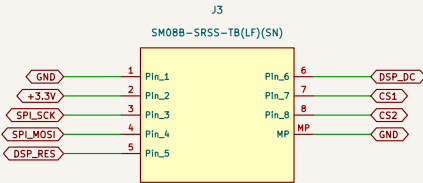
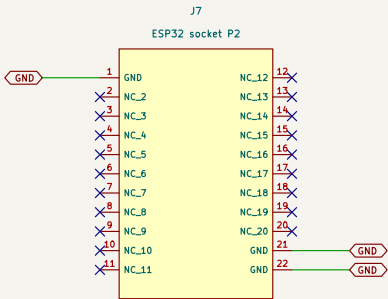
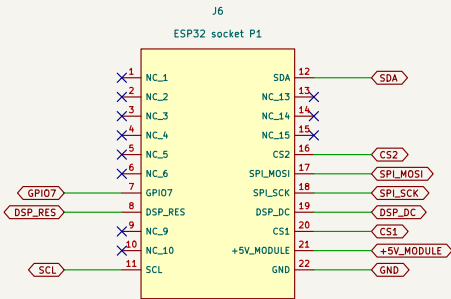
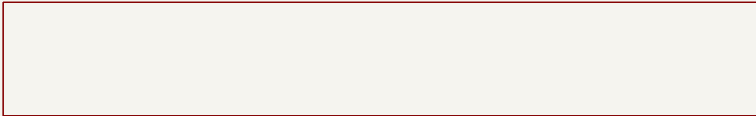


Controller and connectors



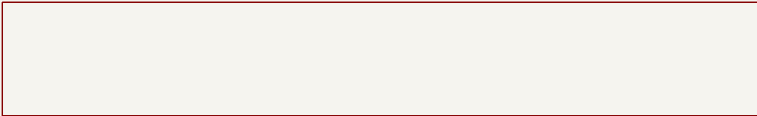
Waveshare 28836 socketed module. ESP 3.3V header pins intentionally isolated.
Peripherals use U7. Disconnect USB before closing J8 for external power.

Accessory protection



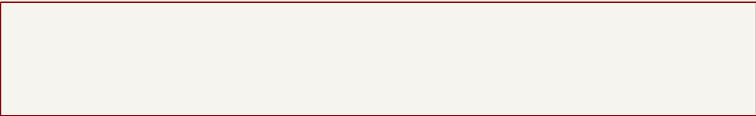
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5V regulator and USB isolation



File: sheet3.kicad_sch

Battery-backed clock and light sensor



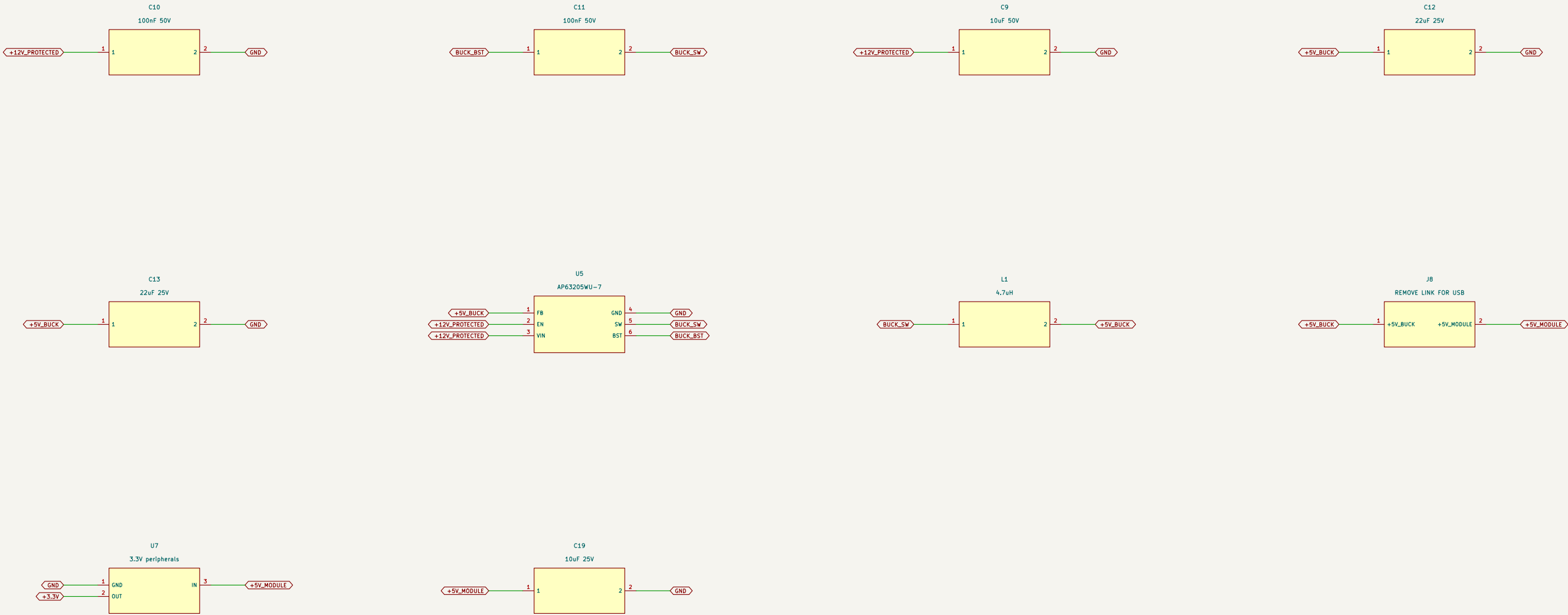
File: sheet4.kicad_sch

Accessory protection



12V variant only: fused accessory input. RTN is isolated from GND.
USB variant: vehicle protection is external to this board.

5V regulator and USB isolation



J8 open for USB. USB unplugged before closing J8.
External power and USB must not be connected simultaneously.
U7 supplies peripheral 3.3V; never bridge it to the ESP 3.3V output.

Battery-backed clock and light sensor

PCF8563 at I2C 0x51. D5/D6 isolate the non-rechargeable CR2032. X1 plus C18 form the 32.768kHz oscillator; verify drift on prototype. Remote BME280 provides temperature/humidity/pressure; no local MCP9808.

Sheet: /Battery-backed clock and light sensor/
File: sheet4.kicad_sch

Title: RykerConnect – Battery-backed clock and light sensor

Size: A3 Date: 2026-09-13 Rev: REV05C-12V
KiCad E.D.A. 10.0.6 Id: 4/4

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Sheet: /Battery-backed clock and light sensor/ File: sheet4.kicad_sch	
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Size: A3	Date: 2026-09-13
KiCad E.D.A. 10.0.6	Rev: REV05C-12V Id: 4/4